

Computation and linguistic meaning: Evaluate, improve, and apply language models in linguistic research

Evaluate

DSMs in Chinese

IMPROVE

Pragmatics

APPLY

L2 Degree Expressions

Computation and linguistic meaning

Study 1

Evaluate LMs in
Chinese word
meaning tasks

Study 2

Finetuning improve
LMs' performance in
pragmatic intent
detection

Study 3

Apply LMs in
tracking
L2 learning of degree
expressions

Evaluate

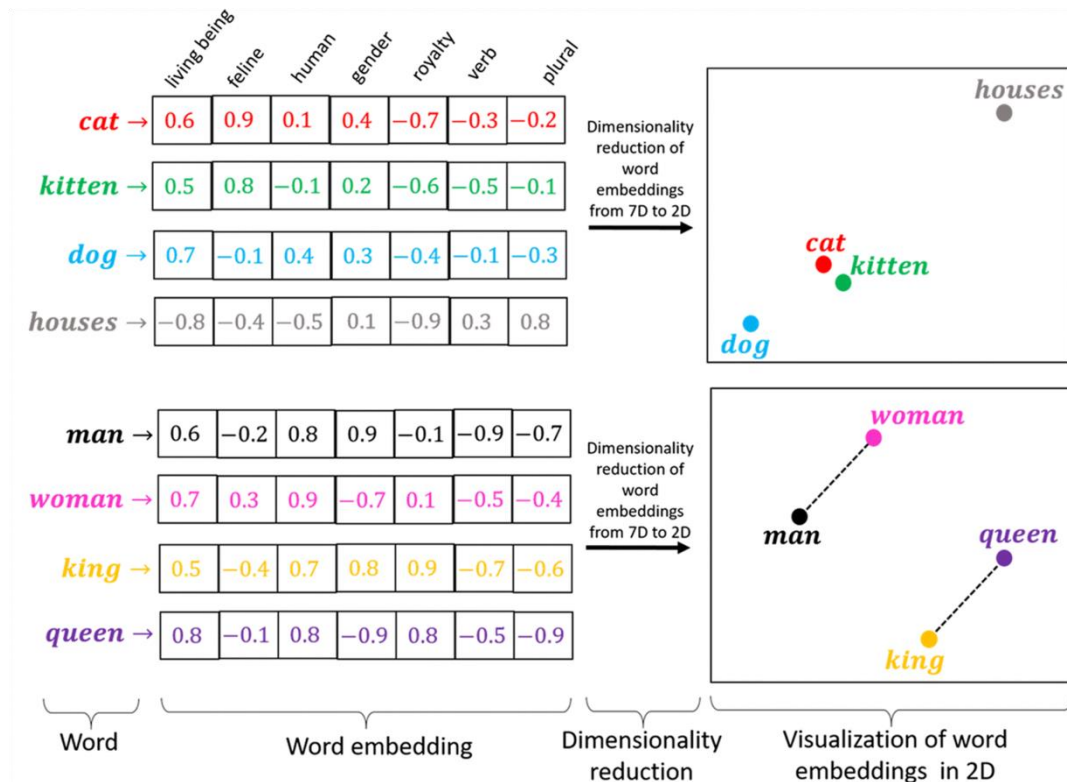
01

Comparing Static and Contextual Embeddings in Chinese

What is Distributional Semantics?

- ❑ Distributional Semantics: assumes that *words* appearing in similar contexts have similar meanings.
- ❑ Distributional Semantic Models (DSMs) are data-driven semantic representations from large text corpora.

What is Distributional Semantics?



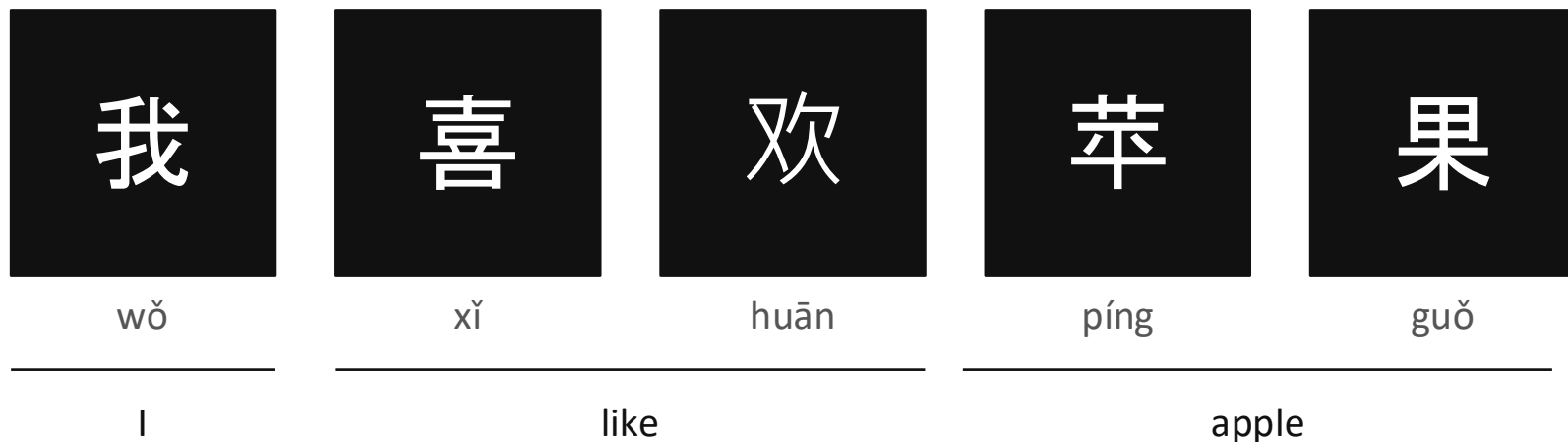
$$\overrightarrow{\text{king}} - \overrightarrow{\text{male}} + \overrightarrow{\text{female}} \approx \overrightarrow{\text{queen}}$$

Firth (1957),
 Jurafsky and Martin (2023),
 Elvevåg et al (2007),
 Solanki (2022)

What is Distributional Semantics?

- ❑ Distributional Semantics: assumes that *words* appearing in similar contexts have similar meanings.
- ❑ Distributional Semantic Models (DSMs) are data-driven semantic representations from large text corpora.
- ❑ *Static* models: a single, global semantic representation for each *word* type (word2vec, skipgram)
- ❑ *Dynamic* models: Contextualized (like ELMO, BERT, GPT)

“Word” boundaries in Chinese - challenging



Transformers tokenize Chinese at character level; no native word representations

→ Reconstructing word representation from character pieces costs model performance

Classic DMS benchmarks are word-level

→ 84%+ of dataset vocabulary are multi-character words

Two paradigms of word meaning representation

STATIC

One vector
per word type

Skip Gram

- Words only
- Words + N-grams
- Words + Characters
- Words + N-grams + Characters

CONTEXTUAL

One vector
per token context

Transformers (12 layers)

- BERT-Base Chinese
- DeBERTa Chinese
- GPT-2 Chinese

Layers: First 4 (F4) / Middle 4 (M4) / Last 4 (L4) / Final

Lenci et al., 2023;
Li et al., 2018;
Qiu et al., 2018.

Five intrinsic tasks on Mandarin Chinese

T1 Similarity Estimation

COS960 · 960 word pairs rated by 30 native speakers

T2 Word Associations

FAST-zh · 300 four-word tuples (stimulus / first / higher / random)

T3 **Semantic Analogies**

CA8 · 7,363 semantic questions, 28 relation types

T4 Semantic Relations

EVALution-MAN · 3,923 pairs: synonymy, hypernymy, antonymy

T5 Semantic Clustering

Binder-zh (535 words, 11 classes) · Zhong22 (664 nouns, abstract/concrete)

Semantic Analogies

中国 : 北京 = 意大利 : ?

China : Beijing = Italy : ? → 罗马 (Rome)

Accurate if

Highest Similarity

[vector[Italy – China + Beijing], vector[Rome]]

$$\overrightarrow{king} - \overrightarrow{male} + \overrightarrow{female} \approx \overrightarrow{queen}$$

Model	cos	ρ
BERT-Base First 4	0.39	0.39
BERT-Base Last	0.84	0.84
BERT-Base Last 4	0.82	0.82
BERT-Base Middle 4	0.67	0.68
DeBERTa First 4	0.44	0.44
DeBERTa Last	0.63	0.63
DeBERTa Last 4	0.63	0.63
DeBERTa Middle 4	0.57	0.57
GPT-2 First 4	0.41	0.41
GPT-2 Last	0.38	0.38
GPT-2 Last 4	0.44	0.43
GPT-2 Middle 4	0.44	0.43
SkipGram N+C	0.93	0.93
SkipGram N	0.97	0.97
SkipGram C	0.91	0.91
SkipGram	0.93	0.92

Table 11: Semantic analogies results for the CA8 dataset. We show accuracy using cosine (*cos*) and Spearman (ρ) as similarity metrics.

Take-away

Static models remain strong for tasks that treat word meaning as context-independent.

Transformers excel at capturing relational structure, context-dependent tasks, semantic classes.

Chinese character tokenization systematically disadvantages transformers on word-level tasks.

No single model wins universally - task structure determines which representation wins.

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Improve

02

Psycholinguistic Diagnosis of LMs' Commonsense Reasoning

Why common-sense pragmatics reasoning matters for AI and LLMs

“For AI models to be truly helpful, they need to understand the *unspoken* rules about how the world works...”

“...AI is incredibly smart and shockingly stupid...”

“...Common sense, in my view, is the dark matter of intelligence and language.”



Yejin Choi (2024)

What LMs miss: pragmatic intent

PRESUPPOSITION

“The guard
spoke to him.”

Presuppose existence of a unique guard.
Plausible context: *Seth went to jail.*
Implausible: *Seth went to a restaurant.*
(Singh et al., 2016)

SCALAR IMPLICATURE

“I ate some
of the cookies.”

some implies not all.
SI can be unnoticed
when a relative clause is present.
(Nieuwland et al., 2010)

Scalar Implicature

So-so

Some people have lungs that require good care.

SI unnoticed

Plausible

Some people have pets that require good care.

Implausible

Some people have lungs, which require good care.

SI noticed

Plausible

Some people have pets, which require good care.

Only pragmatically skilled participants (lower autism-spectrum scores)
show N400 sensitivity to scalar implicature violations

(Nieuwland et al., 2010)

Two LMs, two evaluation strategies

DistilBERT

67M params · Encoder only · Bidirectional
[open-source]

Strategy:

MASK the critical word
(e.g., *jail* vs. *restaurant*).
Success if critical word is in
top-5 ($K=5$) predictions for
plausible sentence, and NOT
in top-5 for implausible.

GPT-3

175B params · Decoder only · Unidirectional
[close-source]

Strategy:

Sequential word prediction.
Sum of token log-probabilities
normalized by sentence length
= proxy for sentence plausibility.
Success if
plausible sentence mean > implausible mean.

ImpPress: fine-tuning data for pragmatics

Total size	>25,000 sentence pairs, semi-automatically generated
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Presupposition training	14,100 tagged pairs · 1,410 test pairs
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SI training	6,000 tagged pairs · 600 test pairs
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Presupposition example input	S1: 'Victoria's mall that has hurt Sam might upset Helen.' S2: 'Victoria doesn't have exactly one mall that has hurt Sam.' · Label: contradiction
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SI example input	S1: 'The teacher resembles some sketches.' S2: 'The teacher doesn't resemble all sketches.' · Label: entailment
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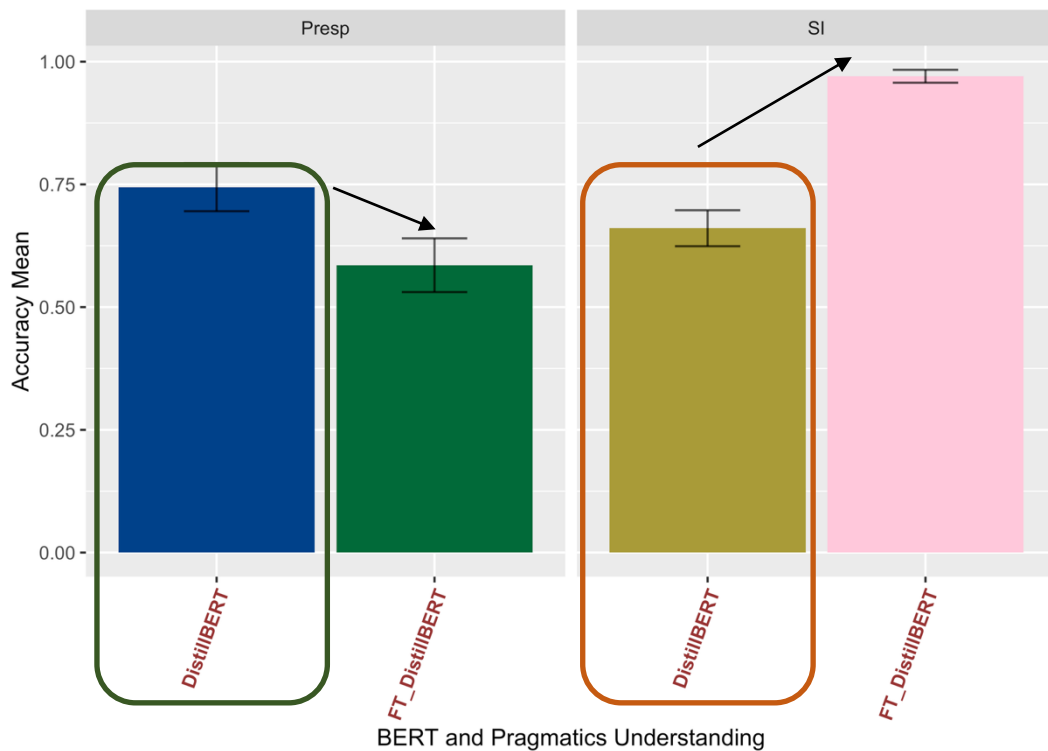


Figure 1: Evaluate BERT with human data. DistillBERT is used for critical word prediction. FT: fine-tuned.

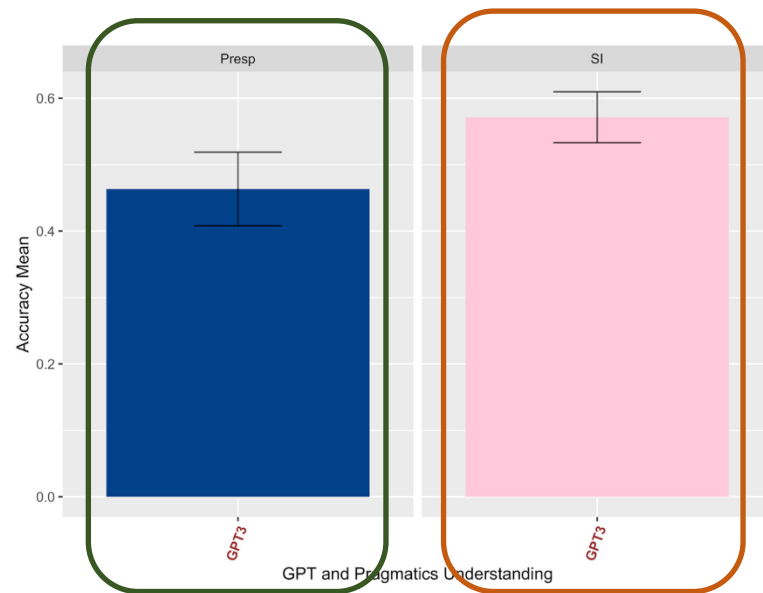


Figure 2: Evaluate GPT-3 with human data. GPT-3 is used for sequential word prediction.

Take-away

DistilBERT seems to have latent presupposition sensitivity even without fine-tuning.

Fine-tuning on linguistically annotated dataset improves implicature detection.

GPT-3 performs near chance – striking findings for a model of that scale.

Psycholinguistic datasets (online + offline) and LM evaluation.

Natural tasks and strategy that match model architecture - BERT $\neq$ GPT.



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Apply

03

L2 Learning of Degree Expressions: A Computational Approach

Degree semantics

"Alex is taller than Kai."

Alex	is tall	-er	than Kai
target	Gradable predicate	Comparative <i>-er</i> morpheme	Standard of comparison

ENGLISH

Particle comparative:
'taller than' / 'more expensive than'

Both phrasal AND clausal standards available

CHINESE

Exceed comparative:
Alex bǐ Kai gāo'
Alex exceeds Kai tall

No overt morphological marking.
Only phrasal standard.

Stassen, 1984;

Kennedy & McNally 2005

Markedness Differential Hypothesis

(Eckman, 1977, 2014; among others)

If A implies B but B does not imply A, then A is more marked than B.

Clausal Comparative
(e.g., 'taller than Kai is')



Phrasal Comparative
(e.g., 'taller than Kai')

MORE MARKED

LESS MARKED

(*arguably*) No language has clausal without phrasal — but Chinese has only phrasal.

Prediction: L1 Chinese learners will avoid English clausal comparatives -
they are both different from L1 AND more marked.

*L1 Chinese learners of English · PELIC corpus***Prediction 1**

Higher surprisal → lower usage
naturalness; lower surprisal → more
native-like use of degree expressions

Prediction 2

Surprisal alone cannot capture
avoidance or underuse - frequency
patterns must be examined separately

Prediction 3

LM surprisals should load onto distinct
factors, relative to classic cohesion and
complexity indices

FINDINGS

- Surprisal decreases as proficiency grows (L3→L4→L5)
- Outperforms classic cohesion indices in classifying proficiency
- Smaller models (DistilGPT-2) most decisive predictor

- Clausal comparatives avoided at all levels - absent in L1 Chinese and more marked (c.f. Eckman 1977)
- Complex forms increase with proficiency

- LMs load onto 2 unique factors (sentence surprisal; target adjective surprisal)
- Classic indices do not enter top 10 predictors
- LLMs + cohesion (AUC 0.61) beats either alone

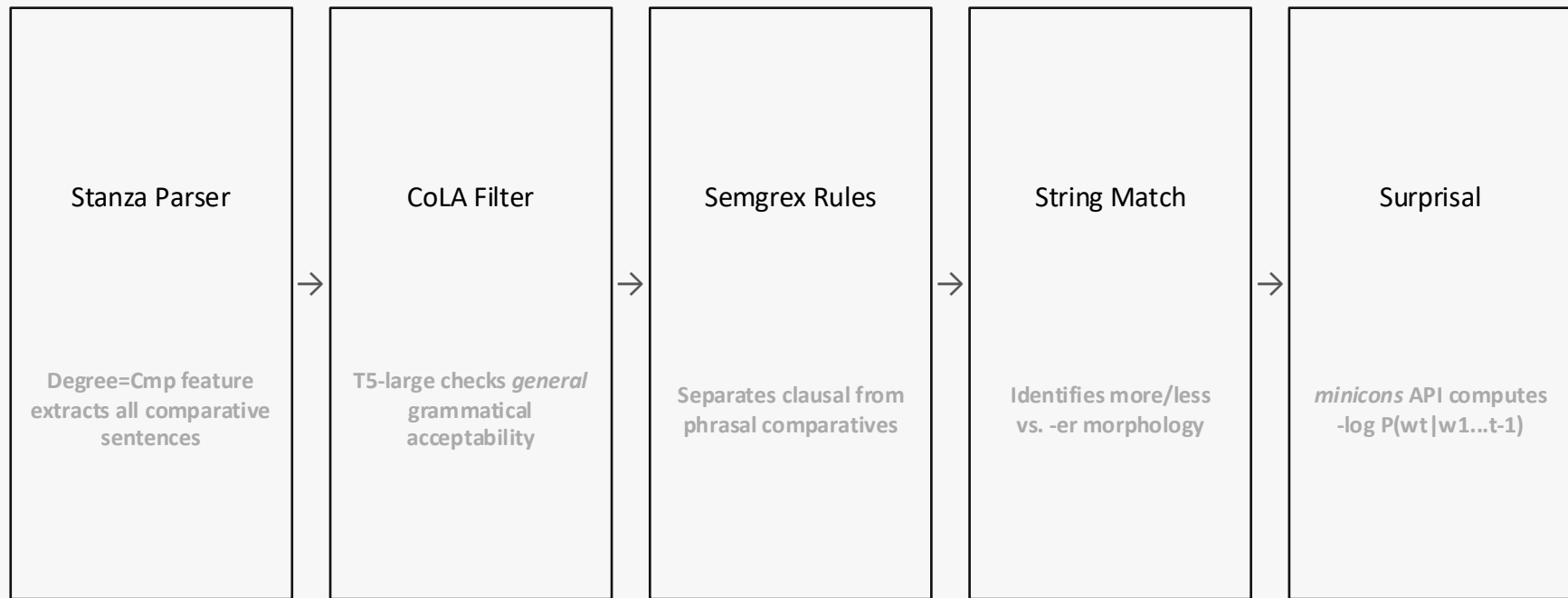
PELIC: a longitudinal learner corpus

Proficiency Level	Learners	Sentences	Cmp Sentences
Level 3	44	4,642	195
Level 4	76	9,081	282
Level 5	48	6,897	235

University of Pittsburgh English Language Institute · 4.2 million words · 7 years · 1,100+ students · Longitudinal

Cmp sentences are rare in learner writing — normalized by total Cmp (Comparative constructions), not total sentences.

Automated extraction pipeline



Degree expression subtypes

SYNTAX	Phrasal_Cmp <i>Alex is taller than Kai.</i>	dependency parse
SYNTAX	Clausal_Cmp <i>Alex is taller than Kai is.</i>	CoLA filter + Semgrep
SYNTAX	Implicit_Cmp <i>Alex is taller.</i>	dependency parse
MORPHOLOGY	More_Cmp <i>This is more expensive than that.</i>	string matching
MORPHOLOGY	ER_Cmp <i>This is longer than that.</i>	string matching

Frequency aspects

Variable	contrast	Estimate	SE	t	p
Intercept		0.41	0.04	9.74	<2e-16
Phrasal_Cmp	Level 3 - Level 4	0.20	0.05	3.91	0.00
	Level 4 - Level 5	-0.003	0.05	-0.07	0.998
Intercept		1.179e-02	7.996e-03	1.47	0.14
Clausal_Cmp	Level 3 - Level 4	0.004	0.01	0.33	0.94
	Level 4 - Level 5	-0.004	0.01	-0.42	0.91
Intercept		0.59	0.04	14.05	<2e-16
Implicit_Cmp	Level 3 - Level 4	-0.2	0.05	-3.91	0.00
	Level 4 - Level 5	0.003	0.05	0.07	0.998
Intercept		0.42	0.04	9.37	5.98e-16
More_Cmp	Level 3 - Level 4	-0.15	0.06	-2.77	0.02
	Level 4 - Level 5	0.03	0.05	0.62	0.81
Intercept		0.58	0.04	13.19	<2e-16
ER_Cmp	Level 3 - Level 4	0.15	0.06	2.77	0.02
	Level 4 - Level 5	-0.03	0.05	-0.62	0.81

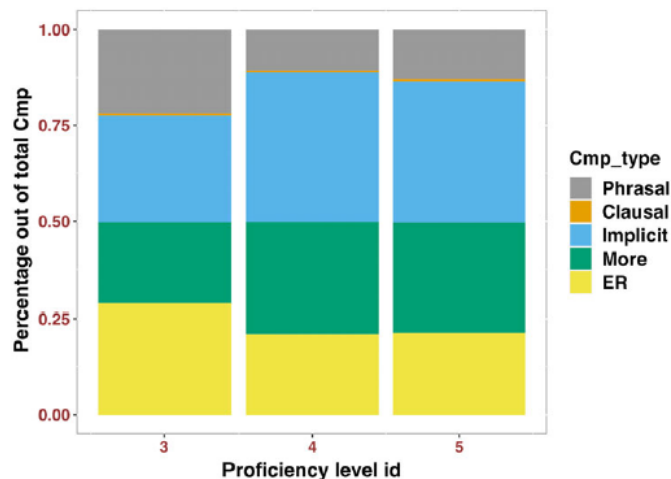


Figure 3. Degree expressions subtypes usage development across proficiency levels.

Table 5.
Linear mixed-effects
models summary for each
degree expression subtype

Three factors for L2 learning of degree expressions

Factor 1	Loading	Factor 2	Loading	Factor 3	Loading
news_av_freq_log	0.92	llama2_sentence_surprisal	0.76	gpt2_target_surprisal	0.73
acad_av_freq_log	0.92	distilgpt2_sentence_surprisal	0.75	distilgpt2_target_surprisal	0.71
mag_av_freq_log	0.91	gptneo_sentence_surprisal	0.75	gptneo_target_surprisal	0.71
all_av_freq_log	0.90	gpt2_sentence_surprisal	0.72	llama2_target_surprisal	0.60
mag_av_freq_type	0.86	gptneo_target_surprisal	0.61	distilgpt2_len_tokens	0.45
fic_av_freq	0.86	gpt2_target_surprisal	0.59	gptneo_len_tokens	0.45
all_av_lemma_freq_log	0.85	distilgpt2_target_surprisal	0.54	MLS	0.44
news_av_freq	0.85	llama2_target_surprisal	0.48		
acad_av_freq_type	0.81	MLT	−0.47		
vac_frequency	0.74	gptneo_len_tokens	−0.72		
frequency	0.68	distilgpt2_len_tokens	−0.72		
diversity_and_frequency	0.62	MLS	−0.73		
acad_av_lemma_freq	0.56				
acad_av_lemma_freq_type	0.56				

Table 7. Factor analysis of the surprisals and existing (syntactic) sophistication indices

LLMs reveal dimensions invisible to classic indices — unexpectedness is a distinct construct.

Three distinct factors L2 learning of degree expressions

F1

Syntactic Complexity

Construction frequency
in COCA sub-corpora
(news, academic, magazine, fiction)
VAC frequency · lemma frequency

TAASSC indices only — LLMs do NOT load here

F2

Sentence Surprisal
& Length

All 5 LLM sentence surprisals
MLT · MLS (mean sentence length)
len_tokens (negative loading)

Longer sentences → lower surprisal

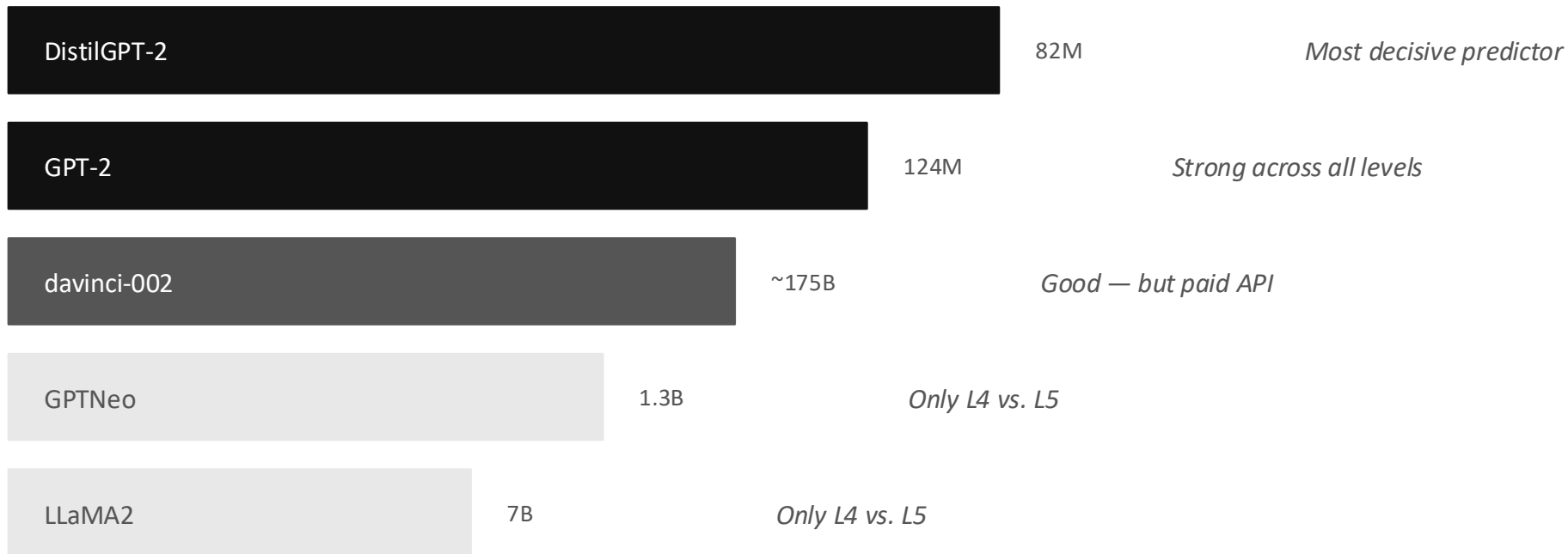
F3

Target Word
Surprisal

GPT-2, DistilGPT-2, GPTNeo,
LLaMA2 target surprisals
Token length (positive loading)

LLMs reveal dimensions invisible to classic indices — unexpectedness is a distinct construct.

Model properties - in L2 learning context



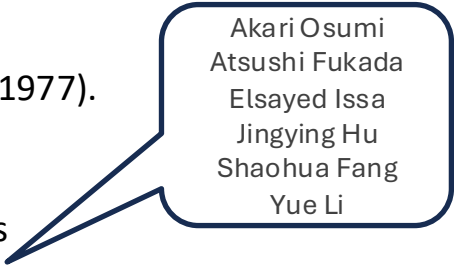
Compression and efficient smaller models can outperform large ones on fine-grained L2 tasks.

Take-away

LLM-surprisals decrease with L2 proficiency in degree expressions usage – a potential developmental metric (also see Cong, 2025).

Clausal comparatives are avoided across all levels – c.f. Eckman (1977).

LLMs capture dimensions of L2 learning *invisible* to classic indices
– not redundant, not a replacement, but complementary.



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Smaller LLMs (DistilGPT-2) outperform larger ones – we do not always need the largest model to get the most meaningful signal for L2 acquisition research.

To conclude

Computation in linguistic research

Evaluate: Static and contextual models reveal complementary aspects of word meaning in Chinese.

Improve: Psycholinguistics and linguistically annotated datasets improves LMs' knowledge about pragmatics.

Apply: LLM-surprisals (together with a frequency model) to track L2 development in theoretically motivated ways.

Implications

Translate and interpret LMs in language studies: Practical recommendations

	DistilBERT 67M	Study2	<i>Best pragmatic sensitivity; fine-tuning works</i>
	DistilGPT-2 82M	Study3	<i>Most decisive feature for L2 classification</i>
	GPT-2 124M	Study1, Study3	<i>Strong analogies (Study1); good L2 tracking (Study3)</i>
	GPTNeo 1.3B	Study3	<i>Only distinguishes adjacent advanced levels</i>
	LLaMA2 7B	Study3	<i>Same pattern - larger ≠ more sensitive</i>
	GPT-3 175B	Study2	<i>Near chance on psycholinguistic tasks</i>

Interpretable with respect to a particular linguistic and second language acquisition theory

Language specific considerations

Study 1

Character-level **tokenization** means transformers have no *native* word representation. Reconstructing word representation from character pieces costs model performance.

Study 2

Implicatures in different languages – **data scarcity** (annotated and validated) [*work in progress*]

Study 3

L1 Chinese learners avoid English clausal comparatives because they are absent in Chinese AND more marked - a **typologically** related learning difficulty

- integrating LLMs with existing, classic models and measures
- more interpretable results

Final thoughts

Can LLM do ...?

- The efficacy of AI is determined by its domain-specific expertise. Language experts define what counts as effective or meaningful language measures in language studies.
e.g., *LLM-indices seem precise and efficient; as a new construct? Open-to-discussion...*

“Human-like” AI?

- There are profound differences between humans and AI, and we need a nuanced perspective when translating AI to study populations like L2 learners
e.g., *combining LLM-indices and basic frequency models could better account for L2 questions.*

Besides automation, what else can we do with LLMs?

- While AI has “surpassed” human capabilities in certain domains and is useful in automating language learning, humans will always have a unique role in evaluation, interpretation, and optimization.

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S1: github.com/yancong222/DSM-Chinese-LLMs · S2: github.com/yancong222/Pragmatics-Commonsense-LMs · S3: github.com/yancong222/NLPDegreeExpression

01

EVALUATE

Chinese DSMs

Static vs. contextual embeddings
on intrinsic tasks

LREC-COLING 2024

02

IMPROVE

Pragmatics

Probing LMs with psycholinguistic
datasets; fine-tuning DistilBERT

CSRR Workshop 2022

03

APPLY

L2 Degree Expressions

LLM-surprisals + frequency
to track L2 development

Natural Language Processing 2025

Surprisal ~ naturalness

$$\text{Surprisal}(w_t) = -\log P(w_t \mid w_{1\dots t-1})$$

Low surprisal

A sequence of strings is expected given context → more natural

High surprisal

A sequence of strings is unexpected → less natural

Usage shifts across proficiency levels



Phrasal_Cmp

L3 → L4 $p < .01$

Learners rely less on simple phrasal forms



Implicit_Cmp

L3 → L4 $p < .01$

Flexibility in comparative expression grows



More_Cmp

L3 → L4 $p < .05$

Complex multi-syllabic adjectives need 'more'



ER_Cmp

L3 → L4 $p < .05$

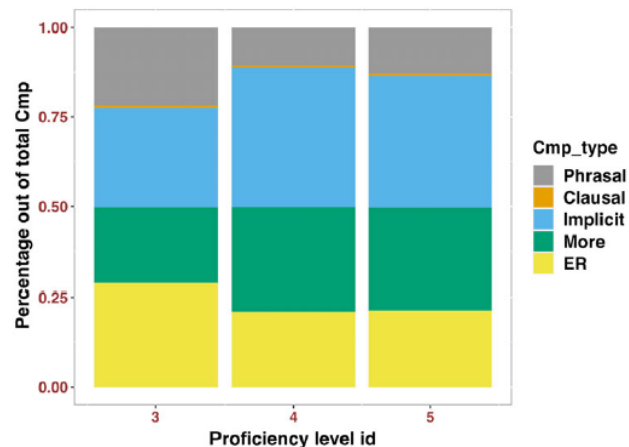
Simple -er adjectives become less dominant



Clausal_Cmp

L3 → L4 rare at all levels

Marked, absent in L1 — strongly avoided



Surprisal: one formula, three interpretations

$$\text{Surprisal}(w_t) = -\log P(w_t \mid w_{1\dots t-1})$$

Study 1

Embedding quality [*extension – work in progress*]

How well does a model represent word meaning in context vs. out of context?

Study 2

Pragmatic sensitivity

Does the model assign lower probability to pragmatically implausible continuations?

Study 3

L2 proficiency

Do more proficient learners produce lower-surprisal degree expressions?

What LLMs reveal about language

Distributional hypothesis

Static embeddings capture type-level word meaning surprisingly well — reinforcing distributional semantics as a valid theory of meaning.

(Gricean) Pragmatics

Some pragmatic inferences (presupposition) are latent in pre-trained representations; scalar implicature requires additional learning from inference data.

Markedness theory

Eckman's 1977 predictions hold computationally: avoidance frequency maps directly onto markedness hierarchy across proficiency levels.

Typological universals

Language-specific properties (Chinese tokenization, comparative typology) systematically affect LLM behavior - LLMs are not language-neutral.

Context sensitivity: task determines winner

Out-of-context meaning
(Analogies, Associations)

→ *Static*

Type-level representation
dominates

In-context meaning
(Semantic Relations, Clustering)

→ *Contextual*

Token-level representation
dominates

Pragmatic sensitivity
(Presupposition, SI)

→ *Contextual LLMs, with fine-tuning*

fine-tuning helps SI

L2 learning tracking
(Degree expressions)

→ *Contextual LLMs*
(smaller = better)

DistilGPT-2 most
decisive predictor

Limitations

S1 Focused on Chinese only; findings may not generalize across language families. SVD dimensionality reduction hurts contextual model performance.

S2 Stimuli are written sentences, not conversational data — yet implicatures are conversational phenomena. Fine-tuning format may mismatch test items.

S3 Writing prompts not controlled for degree expression elicitation. Overall writing score used as proxy for degree expression proficiency. Small corpus for clausal Cmp.

All LLMs are trained on web data (English dominant); their surprisals may not fully capture learner interlanguage. N-gram co-occurrence confounds in psycholinguistic stimuli.

Future / ongoing work

- Extend to other L1 groups (Japanese, Korean, Arabic) and more linguistic constructions
- Conversational corpus evaluation — test pragmatic inference in dialogue settings
- Fine-tune LLMs on learner corpora — can a learner-specific model track development better?
- VLMs (vision-language models) — ongoing work on multimodal pragmatic reasoning
- Multilingual benchmarks that account for tokenization unit differences across languages

Key References — Study 1 & 2

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